

Representative Formal Research Note

Question

This note checks whether a short research report with mixed Chinese and English prose can be rendered as a formal PDF without changing the requested profile identity.

Evidence

第一段说明研究问题：我们希望比较一个简单线性预测器和带正则项的版本。The report intentionally mixes scripts so font pairing, line spacing, and mathematical context are visible in the rendered artifact.

The core objective is

$$\min_{\theta \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^n \ell(y_i, x_i^\top \theta) + \lambda \|\theta\|_2^2.$$

Compact Result Table

Model	Metric	Value	Interpretation
Baseline	RMSE	1.42	reference setting
Regularized	RMSE	1.18	lower is better

Interpretation

The numerical values are synthetic and only serve as a rendering fixture. The paragraph should look like normal body text, not a compressed table export or a browser printout.